

packages/graph-core/src/analyzer/graph-builder.ts

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Overview

A new file `packages/graph-core/src/analyzer/graph-builder.ts` (lines 1-337) introduces a `GraphBuilder` class that assembles a knowledge graph from source files, structural analysis, and non-code artifacts. The file also defines metadata interfaces and a kind-to-node-type map.

Key changes

- **Imports** (lines 1-13): ``ts import type { KnowledgeGraph, GraphNode, GraphEdge, StructuralAnalysis, DefinitionInfo, ServiceInfo, EndpointInfo, StepInfo, ResourceInfo, SectionInfo } from "../types.js"; import { LanguageRegistry } from "../languages/language-registry.js";

``- **Metadata interfaces** (lines 15-37): `FileMeta`, `FileAnalysisMeta`, `NonCodeFileMeta`, `NonCodeFileAnalysisMeta`.

- **Kind mapping** (lines 39-58): `KIND_TO_NODE_TYPE` maps definition kinds (e.g., `"table"`, `"service"`) to graph node types.
- **GraphBuilder class** (lines 60-337):
 - Holds `nodes`, `edges`, `languages`, `nodeIds`, `edgeKeys`.
 - Constructor accepts `projectName`, `gitHash`, optional `LanguageRegistry`.
 - `detectLanguage` uses the registry.
 - File addition: `addFile`, `addFileWithAnalysis` create file nodes and, for the latter, function and class nodes with `"contains"` edges.
 - Edge addition: `addImportEdge` (weight 0.7) and `addCallEdge` (weight 0.8).
 - Non-code handling: `addNonCodeFile`, `addNonCodeFileWithAnalysis` create nodes for definitions, services, endpoints, steps, resources, and link them via `addChildNode`.
 - `addChildNode` warns on duplicate IDs.
 - `mapKindToNodeType` warns on unknown kinds and falls back to `"concept"`.
 - `build` returns a `KnowledgeGraph` with sorted languages and current timestamps.

Impact

- Centralizes graph construction logic, reducing duplication across analyzers.
- Provides explicit metadata contracts via interfaces.
- Uses `Set` structures for duplicate detection, improving correctness.
- No existing public APIs are altered; the new file is isolated.

Risks & follow-ups

- **Language detection:** `detectLanguage` depends on `LanguageRegistry`; verify that all project extensions are registered.

- **Duplicate node handling:** `addChildNode` silently skips duplicates after a warning; confirm downstream consumers tolerate missing nodes.
- **Kind mapping coverage:** `KIND_TO_NODE_TYPE` may miss future kinds; add tests for unknown kinds to ensure fallback to `"concept"`.
- **Edge weight semantics:** Current weights (0.7 for imports, 0.8 for calls) are arbitrary; confirm they align with downstream traversal logic.